

Phase 1: Business Definition

Task 1.1 – Writing the business description outline

COMP433 Software Engineering – Section 5

Semester 2 (Spring) 2025/2026

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Where we are now

Phase 0 closed; Phase 1 starts today

Phase 0

complete

Initiation

Group formation, business preferences, role allocation. Closed Sat 2 May 2026.

Phase 1

now

Business definition

Customer groups write a 1–2 page business description. Workshop with developer group on the deadline date.

Phase 2

to come

Requirements engineering

User requirements, system requirements, effort and cost estimation. All work in developer role.

Phase 3

to come

Analysis and modelling

Use case diagram, use case descriptions, activity diagram.

Phase 4

to come

System modelling and design

Class diagrams, sequence diagrams, design goals, architecture, components, deployment.

Why this phase exists

Software engineering starts before code

“Requirements engineering is the process of establishing the services that the customer requires from a system, and the constraints under which it operates and is developed.”

Sommerville (2016), Software Engineering 10th edition, Chapter 4

In plain language

You cannot design what you do not understand. Most failed software projects fail because the team built the wrong thing, not because they coded it badly.

Before requirements come the business and the people. What does the business actually do? Who uses it? What problems does it have today?

Phase 1 forces you to slow down and look at the business through the customer's eyes. Phase 2 onwards then turns that understanding into concrete requirements.

Two roles per group

Why every group is both customer and developer

Each group has a primary role as developer, and a secondary role as customer. The customer role is used only in Phase 1, Task 1.1. From Phase 2 onwards, every group works only on its own developer business.

How the cycle works

Developer role

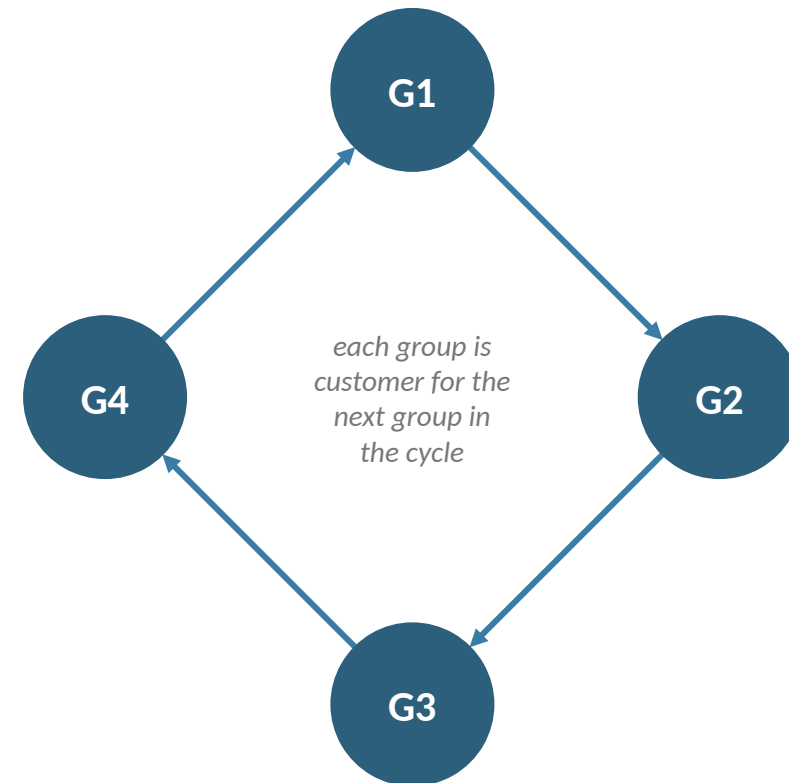
Your group develops your own allocated business across Phases 1 to 4. This is your primary role for the rest of the semester.

Customer role (Task 1.1 only)

Your group writes a business description for the developer group next to you in the cycle. After Phase 1 submission, the customer relationship ends – every group works only on its own business from Phase 2 onwards.

Find your pairing

See the Cycle Summary sheet in the attached spreadsheet – your customer-developer pairing is shown in one row.



What you produce in Task 1.1

A business description outline, one to two pages

What it IS

A description of an existing, real business in the chosen domain – written from the customer's perspective.

Specific, concrete, and grounded in real data: actual numbers, real services, real employees, real processes.

Written in business language, not technical language. A non-programmer should understand every sentence.

One to two pages: enough to be detailed; short enough to stay focused on what matters.

What it IS NOT

A description of the system you would like to build – that is Phase 2's job.

A list of user requirements or features. Phase 1 describes the business, not the software.

A marketing brochure or sales pitch. You are not selling the business; you are documenting it.

Vague or aspirational text. Avoid phrases like “the system will provide a smooth experience” – they say nothing.

A clear test: if a stranger reads your description, can they explain the business back to you correctly? If yes, it works. If they leave with questions, it does not.

The five questions

How to structure your business description

Use these five questions as your structure

Each maps to a section of your description

1

What is the business about?

What it is, what domain, what kind of customers it serves

2

What services does the business offer?

List each service explicitly: selling, delivery, payment, support, etc.

3

What is the business capacity?

Real numbers: monthly customers, products, employees, branches

4

How does the business work?

Business processes: a step-by-step flow for each service, the business model

5

Who works there and what do they do?

Employees and their roles, mapped to the services they support

Question 1: What is the business about?

Set the scene in one short paragraph

Aim for a clear, one-paragraph description that includes:

- What kind of business is it (sector, domain)?
- Who are its primary customers (individuals, organisations, both)?
- Where does it operate (single location, branches, online, region)?
- What problem does it solve, or what need does it meet?

Example – for a cake shop business

“The business is a cake shop with three branches in Ramallah, Nablus, and Hebron. It produces and sells cakes, sweets, and pastries for everyday purchase and for occasions such as birthdays, weddings, and graduations. Customers are mostly individuals and families ordering for personal events; a smaller share comes from corporate orders. Today, ordering is done by phone or in person – there is no centralised online ordering system, which makes it difficult to coordinate orders across branches and respond to customers consistently.”

Notice: domain (cake shop), customers (individuals, families, some corporate), where (three named branches), and the problem (no central system). All in one paragraph.

Question 2: What services does the business offer?

Be specific. List each service. Describe each one.

List every distinct service the business provides. For each one: a short label, then 1–2 sentences describing what it actually involves in the existing business (not what the software will do).

Standard cake selection and ordering

Customers choose from displayed options by flavour, size, and design; orders are recorded by hand or by phone.

Custom cake orders

Customers describe a custom cake; the shop reviews feasibility, agrees a price, and confirms before preparation.

Payment

Cash or card, paid in-store or upon delivery depending on order type.

Delivery and pickup

Customers choose between in-store collection and home delivery within the branch's service area.

Customer support

Customers contact the shop to ask about order status, request changes, or raise complaints.

Promotions and discounts

Seasonal offers communicated in-store and via WhatsApp; pricing varies for occasions and seasons.

Question 3: What is the capacity of the business?

Real numbers, not “a lot” or “many”

Cover at least these dimensions, with concrete numbers:

- How many customers does the business serve per month? Per week?
- How many products or services are offered, in how many categories?
- How many concurrent customers or orders are typical at peak time?
- How many branches, locations, or service areas?

210

orders per month
across 3 branches

12

concurrent active orders
at peak time

35

distinct product types
in the catalogue

22

employees in total
(20 branch + 2 central)

Then break the totals down per branch or per category

“Ramallah branch: 8 staff, 85 orders/month, up to 5 concurrent orders. Nablus branch: 6 staff, 75 orders/month. Hebron branch: 6 staff, 50 orders/month. Central staff: 2.”

Question 4: How does the business work?

One business process per service

For each service from Question 2, write a short step-by-step business process – how that service actually works today, before any software. Then describe the business model: how the business makes money.

Example process – “Standard cake order”

- 1 Customer contacts the shop by phone, WhatsApp, or in person, and asks about available cakes (size, flavour, occasion).
- 2 Staff describes options; customer specifies their choice, delivery address, and preferred date.
- 3 Staff confirms verbally and records the order manually in the daily log.
- 4 Kitchen prepares the cake; decorator finishes; driver delivers, or customer collects.
- 5 Customer pays in cash on delivery or in-store; no formal receipt is issued.

Business model – how revenue is made

Main income: selling standard cakes from the catalogue.

Secondary income: custom design fees (depending on complexity), delivery fees (by city), seasonal premium pricing for major occasions.

Question 5: Employees and their roles

Map roles to the services they support

Describe each role in the business: name of the role, how many people hold it, and which service(s) they primarily support. The mapping from roles to services matters – it tells the developer who the system's eventual users will be.

Branch Manager	<i>1 per branch</i>	Daily operations, staff scheduling, inventory
Pastry Chefs	<i>2 per branch</i>	Cake production, quality control
Decorator	<i>1 per branch</i>	Custom cake decoration
Cashier / Customer Service	<i>1 per branch</i>	In-store orders, customer enquiries
Delivery Driver	<i>1–2 per branch</i>	Home delivery within service area
Orders Coordinator (central)	<i>1 total</i>	Cross-branch coordination, monitoring
Promotions Coordinator (central)	<i>1 total</i>	Seasonal offers and discounts

Total workforce: 22 (20 branch staff + 2 central). Be explicit about the total at the end.

What good and weak look like

Two anonymised submissions and the actual instructor feedback

A strong submission

Cake shop business – anonymised from a previous semester

Overall grade: A – needs improvements

What this group did well

- ✓ **Team page.** Names, IDs, group number, AND roles all listed on a separate front page.
- ✓ **Real numbers.** 210 orders/month, 35 product types, 22 employees, broken down per branch.
- ✓ **Listed services.** Six distinct named services, each with a one-paragraph description.
- ✓ **Per-service process.** A step-by-step business process for EACH service, not just the business as a whole.
- ✓ **Role-to-service map.** Each role explicitly tied to the services it supports.
- ✓ **Business model.** Main and secondary income sources clearly identified.

What the marker said about the strong example

Even A-grade work has improvement points

Direct feedback from the marker, paraphrased:

On services

- ✓ Good description of existing business services.
- △ Be specific and direct. Describe how each service actually works in the existing business, not how you would like it to work in the future system.

On capacity

- ✓ Reasonable statistics: orders, monthly customers, employees.
- △ Be more specific on concurrent numbers – how many customers at the same time, not just totals.

On business processes

- ✓ Good business processes for the major services.
- △ Describe the business process for EACH service, not the user's experience. There should be a process per service.

On employees

- ✓ Good description of business employees and counts.
- △ Include the role each employee fills and which business service they support.

On challenges

- ✓ Reasonable.
- △ Describe the business challenges and problems faced by your business – these will help shape Phase 2.

A weak submission

School registration business – anonymised

Overall grade: D+ – poor, needs improvements

What this group missed

- ✗ **Roles missing on the team page.** Names and IDs were listed but each member's project role was omitted.
- ✗ **Services described as system features.** “Allow students to view fees, schedule, calendar.” These are SOFTWARE features, not BUSINESS services. The business service is education and registration.
- ✗ **Vague capacity.** “More than 700 students annually.” No breakdown by grade, no concurrent figures, no class capacities, no monthly numbers.
- ✗ **Process described the future system, not the existing business.** “Parent creates an account, system checks seats...” The marker's note: 'Does your business have a system? Then why do you need development? Describe the existing business process.'
- ✗ **Employees not mapped to services.** “40 staff including teachers, admin, accountant.” No counts per role, no mapping to which service each role supports.
- ✗ **Business challenges section missing entirely.** Nothing about what is hard about running this business today – so the developer has no problem to solve.

Common pitfalls - and how to avoid them

Patterns the marker calls out repeatedly

Pitfall: Describing the system, not the business

Fix: Use present-tense language about current operations. “Customers pay in cash on delivery.” Not “The system shall process payments.”

Pitfall: Vague numbers (“many”, “a lot”, “medium-sized”)

Fix: Replace every vague quantifier with a concrete figure or a labelled estimate. If you must estimate, state your basis.

Pitfall: Total customers but no concurrent figure

Fix: Always include both: customers/orders per month AND peak concurrent customers/orders. They are different and both matter.

Pitfall: One business process for the whole business

Fix: Write one process per service. They share the customer at the start, but they diverge – different staff, different steps, different outputs.

Pitfall: Listing employees without mapping to services

Fix: For each role: count + which services they support. The role becomes an actor in Phase 3 – make it explicit now.

Pitfall: Marketing language

Fix: Avoid “smooth experience”, “user-friendly”, “innovative”. Replace with measurable, observable facts.

The workshop

What happens on the deadline date

Saturday 9 May 2026 in class

The customer-developer interview workshop

Phase 1 ends with a workshop in class. Each customer group meets its developer group, hands over the written description, and is interviewed about it. Then they swap roles in the next pairing.

Before

Submit the written description electronically and bring a paper copy to the lecture. Mandatory paper copy.

Step 1

Customer presents the business description (5 minutes). Developer takes notes.

Step 2

Developer asks follow-up questions (10 minutes): clarifications, missing details, contradictions.

Step 3

Customer answers and notes points to revise (5 minutes).

After

Developer keeps the description as input for Phase 2. Customer's job ends here.

Groups that fail to bring a paper copy to the workshop will be marked down – the workshop cannot work without one.

Submission and timeline

Indicative – confirm with the announcement

Indicative timeline

From today

Phase 1 work begins. Use class time and group meetings.

Wed 6 May

Group L confirms its business choice. Unassigned students contact the instructor.

Fri 8 May

Latest sensible point to finish the draft. Allow time to read it back as a stranger.

Sat 9 May

Submission deadline. Workshop in class. Bring a paper copy.

After 9 May

Phase 2 begins. Customer/developer relationship dissolves; groups work on their own business.

What to submit

1

Cover page. Group letter, member names, IDs, and roles. Customer group → Developer group → Business number/name.

2

Description. 1–2 pages, structured by the five questions. Plain prose with section headings.

3

PDF or Word. Submit electronically using the template provided in the announcement.

4

Paper copy. Bring one printed copy to the lecture on the deadline date for the workshop.

5

Email copy to developer. After submission, also send your developer group a copy by email.

Materials and resources

Where to find the supporting files

All materials for Phase 1 will be shared by the instructor. Use the placeholders below to record the actual links once they are posted on the course page.

Phase 1 announcement

The full announcement explaining the task, deadline, allocations, and group-specific notes.

[announcement link]

Group allocations spreadsheet

Cycle Summary, Project Groups, Unassigned, and Project Timeline. Includes business catalogue 38–52.

[spreadsheet link]

Phase 1 description template

Structured Word template covering the five questions, with cover page formatting.

[template link]

Phase 1 examples (anonymised)

The strong and weak submissions discussed in this lecture, with marker annotations.

[examples link]

Five things to take away

If nothing else, remember these

1

You are describing an EXISTING BUSINESS in your customer role, not a future system. Present tense, real numbers, real people.

2

Use the five questions as your structure: what, services, capacity, processes, employees. Section headings help.

3

One business process per service, not one for the whole business. Map every employee role to the services they support.

4

Specificity is the difference between A and D. Concrete numbers, named services, explicit steps.

5

Submit by Saturday 9 May, bring a paper copy, and email a copy to your developer group.

References

Sommerville, I. (2016). Software Engineering, 10th edition, Chapter 4 (Requirements Engineering). Pearson. (R2)

Bruegge, B. and Dutoit, A. (2013). Object-Oriented Software Engineering Using UML, Patterns, and Java, 3rd edition, Chapter 4 (Requirements Elicitation). Prentice Hall. (R1)

Pressman, R. and Maxim, B. (2020). Software Engineering: A Practitioner's Approach, 9th edition, Chapter 8 (Understanding Requirements). McGraw-Hill. (R5)

IEEE Computer Society (2024). SWEBOK V4 – Knowledge Area: Software Requirements.

Boehm, B. (1981). Software Engineering Economics. Prentice Hall. – for the cost of late requirements changes.

COMP433 Project Structure and Tasks document (Phase 1 Task 1.1 specification).

Phase 1 announcement, Section 5 (Spring 2025/2026).

Anonymised Phase 1 submissions and marker annotations from previous semesters, used for the strong and weak examples.